**Introduction**
Entity Relationship (ER) model proposed by Peter Chen in 1976 is a conceptual representation of data. It is capable of describing the data requirements for a new information system in a direct and easy to understand graphical notation. ER data model represents the overall logical structure of the
DB. ER diagram is a graphical representation of the logical structure of a database.

**Basic ER constructs**

The ER model is designed with the help of the below basic constructs:

- Entity
- Attribute
- Relationship

**Entity and entity types**

An entity is an instance of a physical object in the real world.Represents the real world objects (eg., facts , things , people.,,,etc) that have properties in common.
It is an object that exists and is distinguishable from other objects.
An entity may be

- concrete (eg., person,book, employee etc...)
- abstract (eg., leave,transaction etc...)

Entities have attributes.
Graphically an entity is represented by a rectangle.




**Weak Entity**

- Weak entity is an entity whose existence depends on existence of another entity.
- A weak entity cannot be uniquely identified by its attributes alone, hence it must use a foreign key along with its attributes which is a primary key of the entity on which the weak entity depends upon.
- Graphically weak entity is represented by double rectangle.



**Relationship and relationship types**

A relationship is a logical link between entities.
A relationship can be graphically represented by a diamond.




A relationship can be
a) recursive
b) binary
c) n array
d) have attributes

**Recursive relationship**
A recursive relationship is the one which involves only one single entity. The above representation specifies , a person marries another person.




**Binary relationship**
A relationship which involves two entities. The above representation specifies that 'books' are available in the




'library'.

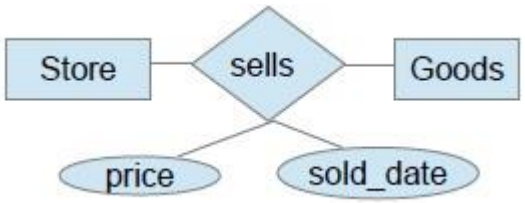
The above representation specifies that 'books' are available in the 'library'.

**Ternary Relationship**
A relationship which involves three entities.



The above ERD represents a ternary relationship.

**Relationship with attributes**
A relationship can also contain an attributes.

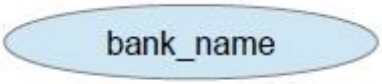



The above ERD represents 'Store sells Items and maintains the quantity sold and the date Item was sold for all the Items sold'.

**ER ATTRIBUTE & CARDINALITIES:**

**Attribute and attribute types**

- An attribute is a property / feature of an entity.
- Attributes are nothing but properties of entities and relationships that describe their characteristics.
- The value of an attribute is an element of a data type like number, date etc..
- An attribute is graphically represented by an ellipse.




**Simple attribute**

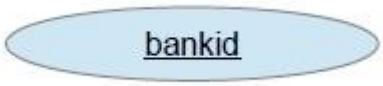
An attribute which can have atomic values, which cannot be divided further.

For example, a employee's phone number is an atomic value of 10 digits.

**Key attribute**

An attribute or a collection of attributes of an entity that uniquely identifies an entity among entity set

- Key attribute is used to represent the primary key.
- Key attribute uniquely identifies the entity.
- Key attribute is graphically represented by ellipse with an attribute name underlined.




**Composite attribute**

An attribute which consists of several simple attributes.

For example : Address consists of 3 attribute Street,city,postal code.

**Derived attribute**

An attribute that do not exist in the physical database, but their values are derived from other attributes present in the database. For example, age of an employee. Age of an employee need not have to store and can be derived from data_of_birth.

Derived attribute is graphically represented as **dashed ellipse**.




**Multivalued attribute**

An attribute with a set of possible values for the same entity. (Eg. Phone no - can have mobile no , land number).A multivalued attribute is graphically represented by double ellipse.

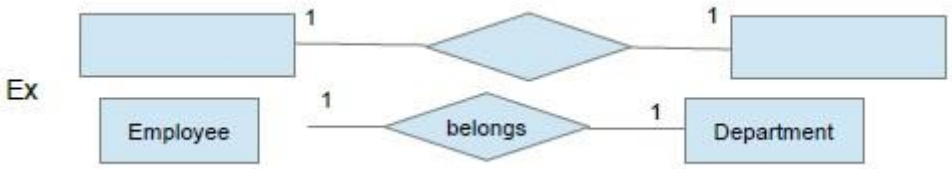



**Cardinalities**

In case of a relationship between two entity sets , cardinality defines the number of entities in one entity set, which can be associated with the number of entities of other set via relationship set..

4 types of cardinalities can be maintained:

a) 1 : 1 ( ONE to ONE )




The above representation indicates “one employee belongs to only one department”.

b) 1 : * ( ONE to MANY )




The above representation indicates “One bank deals with many customers”.

c) * : 1 ( MANY to ONE )




Ex:




d) * : * ( MANY to MANY )




Ex




The above representation indicates 'many stores sells many goods'. Note :( * ) specifies many